MediLink RAG

A Bilingual (Arabic / English) Medical RAG System

Final Project Discussion · 2026-06-11

Retrieval-Augmented Generation · Hybrid Retrieval · Cost-Aware Orchestration · Honest Evaluation

The Problem

- Medical Q&A in **Arabic** is poorly served by general LLMs
- Medical answers demand **factual precision** — a wrong answer causes harm
- LLMs **hallucinate** from parametric memory

Our approach: constrain the model to answer **only from retrieved, verifiable passages** — Retrieval-Augmented Generation.

Two Workloads, One System

Workload	Source	Pipeline
General knowledge "symptoms of diabetes?"	Medical encyclopaedia (3,881 passages)	**Global RAG**
Patient records "when is my next appointment?"	Live patient database	**TOON** (3-tier, cost-aware)

Global RAG Pipeline

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Query (AR/EN)
  → Emergency screening (30+ keywords)
  → Arabic normalisation
  → Query expansion (medical dictionary)
  → Dense (BGE-M3, FAISS)  +  BM25 (translated query)
  → Hybrid RRF fusion (intent-weighted, k=60)
  → Cross-encoder rerank (translate → English first!)
  → Generation (Llama-3.1-8B, Groq)
  → Judge (Gemini 2.0-flash): grounding / hallucination
```

TOON — Token-Optimised Orchestration

Tier	Strategy	Budget	Use case
T1	BM25 + rerank	50 tok	Single-field facts
T2	Dense+BM25 → RRF → rerank → date boost	200 tok	Aggregations
T3	Full patient context + grounding gate	20,000 tok	Reasoning

Key fixes: stable document IDs (was random UUID), date-aware rerank for appointments, token-budget enforcement.

Models

Model	Role
BAAI/bge-m3 (1024-dim)	Dense embedding (AR + EN)
BAAI/bge-reranker-v2-m3	Cross-encoder reranking
Llama-3.1-8B-Instant (Groq)	Generation + translation
Gemini 2.0-flash (Google)	Independent judge

Generator ≠ Judge → removes **same-family agreement bias**.

The Honest Part

We found 3 overfitting / validity threats

No model-weight overfitting (no fine-tuning) — but the evaluation itself was compromised.

Threat 1 — Circular Ground Truth (severe)

The original ground truth = **the retriever's own top-10 outputs**.

> Measuring against it asks "*can you reproduce yourself?*" — not "*are you correct?*"

- Dense recall@10 looked like **0.97**, hybrid MRR **0.99** — implausible
- **Fixed:** new GT builder — independent cosine pool + **Gemini judge**, stable content IDs, no artificial cap
- Honest re-measurement **pending GPU run** → excluded from headline numbers

Threats 2 & 3

Threat 2 — Tuning on the eval set - Pool-size and fusion-weight swept against the *same* set used to report results - Reported gains are an **upper bound**; disclosed as fitted

Threat 3 — Distribution mismatch - Real traffic = 100% general-knowledge queries - Eval set = 60%+ patient-record lookups - The router's cheap T1 path **fires on 0% of real traffic**

Result — Retrieval Ablation (n=100, 95% CI)

Variant	MRR	95% CI	Significant?
BM25 only	0.369	[0.291, 0.453]	✓
Dense only	0.624	[0.545, 0.707]	✓
Hybrid, no rerank	0.400	[0.334, 0.466]	✓
Hybrid, no router	0.702	[0.628, 0.774]	✗
Full system	**0.702**	**[0.628, 0.774]**	—

Two Central Findings

✅ **Cross-encoder rerank is the dominant win** `0.400 → 0.702 MRR (+0.302, p < 0.001)`

❌ **The router adds ZERO retrieval quality** `full = no-router (Δ = 0.000, p = 1.000)` → its value is **cost**, and that cost benefit is unverified on real traffic.

Result — TOON Per-Tier (n=100)

Tier	n	Hit@1	Hit@10	MRR
T1	45	0.578	0.889	0.663
T2	25	0.600	0.920	0.716
T3	30	0.667	0.900	0.751

Small n per tier — one query ≈ 0.04 MRR. Directional, not definitive.

Result — TOON by Category

Strong	Hit@1		Weak	Hit@1
Vitals	0.923		Billing	0.500
Meds	0.857		Diagnosis	0.364
Records	0.636		**Appointments**	**0.357**

Appointments are weakest — they need date arithmetic the reranker can't do.

Result — Router & Grounding

Router accuracy: 94% (conservative errors — over-spend, never under-retrieve)

Slice	Grounded	Hallucination
Overall	0.810	0.160
T2	1.000	0.000
T3	**0.657**	**0.343**

T3 reasons over the full record → highest hallucination → grounding gate needed.

What We Can / Cannot Claim

Claim	Verdict
Reranker is the main retrieval win	✅ measured, significant
Router accuracy 94%	✅ (eval distribution only)
Dense recall@10 = 0.97	❌ circular-GT artifact
Router improves retrieval quality	❌ $\Delta = 0.000$
System is clinically safe	❌ no clinician validation

Limitations

- Honest Global-RAG retrieval metrics **not yet re-measured**
- All TOON results: **n < 50 per tier** → wide CIs
- Eval distribution ≠ real traffic
- Relevance & grounding judged by an **LLM, not clinicians**
- T3 hallucination **0.343** — too high for deployment

Conclusion

- A credible, well-engineered bilingual RAG prototype
- **Strong:** correctly-applied cross-encoder reranking (+0.302 MRR)
- **Honest negatives:** router cost-benefit unverified, T3 hallucination, circular GT
- The real contribution is **diagnosing and disclosing our own evaluation flaws**

> Not clinically ready — but every number is defensible or clearly caveated.

Future Work

1. Complete the **honest retrieval re-measurement**
2. **Expand** the eval set and align it with real traffic
3. Strengthen **temporal reasoning** (appointments)
4. **Calibrate** the Tier-3 grounding gate
5. Obtain **clinician-annotated** gold labels

Thank You

Questions & Discussion

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